Multivariable Calculus

Unit 5
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5 Vector Calculus

5.1 Vector Fields

5.2 Line Integrals

5.3 Fundamental Theorem of Line Integrals

Recall the **Fundamental Theorem of Calculus**, which says that the integral of a function's derivative is the difference of the function's values at the endpoints:

$$\int_a^b f'(x)\,\mathrm{d}x = f(b) - f(a)$$

We can generalize this theorem by noticing that:

- the *integral* is the *sum of little changes* from a to b
  - each *little change* in f is the differential $\mathrm{d}f = f'(x)\,\mathrm{d}x$
- $f(b) - f(a)$ is the *total change* in f from a to b, which we can call Δf
- so, the sum of little changes in f from a to b is equal to the total change in f from a to b:

$$\int_a^b \mathrm{d}f = \Delta f = f(b) - f(a).$$

The fundamental theorem would imply that given some function f, the sum of little changes in f over some curve is equal to the total change in f from the start of the curve to the end.

We know the total differential of $f(x, y, z)$ is $\mathrm{d}f = \frac{\partial f}{\partial x}\,\mathrm{d}x + \frac{\partial f}{\partial y}\,\mathrm{d}y + \frac{\partial f}{\partial z}\,\mathrm{d}z$. Each total differential $\mathrm{d}f$ can be interpreted as a "little change in f". If $\vec{p}$ and $\vec{q}$ are the start and end points of some curve C, then by the fundamental theorem, we can say:

$$\int_C \mathrm{d}f = \Delta f = f(\vec{q}) - f(\vec{p}).$$

Realize that the $\mathrm{d}f = \frac{\partial f}{\partial x}\,\mathrm{d}x + \frac{\partial f}{\partial y}\,\mathrm{d}y + \frac{\partial f}{\partial z}\,\mathrm{d}z$ can be represented as the dot product $\vec{\nabla} f \cdot \mathrm{d}\vec{r}$, since $\vec{\nabla} f = \left\langle \frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}, \frac{\partial f}{\partial z} \right\rangle$ and $\mathrm{d}\vec{r} = \langle \mathrm{d}x, \mathrm{d}y, \mathrm{d}z \rangle$. Thus, this gives us the form:

> **Fundamental Theorem of Calculus for Line Integrals**
>
> Let C be a curve that starts at displacement $\vec{p}$ and ends at displacement $\vec{q}$, and f is a function such that $\vec{\nabla} f$ is continuous on C. Then:
>
> $$\int_C \vec{\nabla} f \cdot \mathrm{d}\vec{r} = f(\vec{q}) - f(\vec{p})$$
>
> for any function $f(\vec{r})$.

We can think of $\vec{\nabla} f$ as the "little changes in f" and integrating over $\vec{\nabla} f$ over the curve C gives us the net change of f over the boundary of C, which are the endpoints.

If C is parameterized using t, then since $\mathrm{d}\vec{r} = \vec{r}'(t)\,\mathrm{d}t$, we can write:

$$\int_a^b \vec{\nabla} f(\vec{r}(t)) \cdot \vec{r}'(t)\,\mathrm{d}t = f(\vec{r}(b)) - f(\vec{r}(a)).$$

Note that if C is piecewise-defined, even though the integral over C is improper, the gradient theorem would hold due to properties of integrals (we can split the integral apart).

Conservative Vector Fields

The gradient theorem tells us that the line integral of $\vec{\nabla} f$ along two different paths is equal, as long as the paths start and end in the same position. In other words, if C_1 and C_2 have the same initial and terminal points, then:

$$\int_{C_1} \vec{\nabla} f \cdot \mathrm{d}\vec{r} = \int_{C_2} \vec{\nabla} f \cdot \mathrm{d}\vec{r}.$$

In other words, if a vector field $\mathbf{F}$ can be represented as the gradient of some function f (if there exists an f such that $\mathbf{F} = \vec{\nabla} f$), then the line integral $\int_C \mathbf{F} \cdot \mathrm{d}\vec{r} = \int_C \vec{\nabla} f \cdot \mathrm{d}\vec{r}$ is independent of path and only depends on the initial and terminal points of C.

Vector fields which meet this condition ($\mathbf{F} = \vec{\nabla} f$) are known as **conservative vector fields**.

5.4 Green's Theorem

Often, line integrals can be difficult to compute. **Green's Theorem** states that a line integral can be evaluated as the result of a double integral.

Green's Theorem

Let D be the region bounded by the curve C and $\mathbf{F} = \langle F_x, F_y \rangle$. Then:

$$\oint_C \mathbf{F} \cdot \mathrm{d}\mathbf{r} = \oint_C \left(F_x\,\mathrm{d}x + F_y\,\mathrm{d}y\right) = \iint_D \left(\frac{\partial F_y}{\partial x} - \frac{\partial F_x}{\partial y}\right) \mathrm{d}A$$

if C is **simple, closed, and positively-oriented**.

Constraints of Green's Theorem

- **Simple closed curves.** Recall that *simple* means the curve does not cross itself. *Closed* means the curve will start and end at the same point.
- **Orientation.** The definition above applies only for *positively oriented* curves, i.e. those that move counterclockwise. If you followed the curve, its interior should be to your left. If the curve is *negatively oriented*, then we must negate the sign of the double integral:

$$\oint_C \mathbf{F} \cdot \mathrm{d}\mathbf{r} = -\iint_D \left(\frac{\partial F_y}{\partial x} - \frac{\partial F_x}{\partial y}\right) \mathrm{d}A$$

if C is **simple, closed, and <u>negatively</u>-oriented**.

5.5 Divergence and Curl

Recall

Divergence of a Vector Field

The divergence of a vector field $\mathbf{F}(x, y, z) = \langle F_x, F_y, F_z \rangle$ is:

$$\operatorname{div} \mathbf{F} = \vec{\nabla} \cdot \mathbf{F} = \frac{\partial F_x}{\partial x} + \frac{\partial F_y}{\partial y} + \frac{\partial F_z}{\partial z}.$$

The notation $\vec{\nabla} \cdot \mathbf{F}$ can be used to represent divergence, treating $\vec{\nabla}$ as a vector of partial derivative operators:

$$\vec{\nabla} \cdot \mathbf{F} = \left\langle \frac{\partial}{\partial x}, \frac{\partial}{\partial y}, \frac{\partial}{\partial z} \right\rangle \cdot \langle F_x, F_y, F_z \rangle = \frac{\partial}{\partial x} F_x + \frac{\partial}{\partial y} F_y + \frac{\partial}{\partial z}.$$

We will use this definition of $\vec{\nabla}$ to derive the *curl* of a vector field:

Curl of a Vector Field

The curl of a vector field $\mathbf{F}(x, y, z) = \langle F_x, F_y, F_z \rangle$ is:

$$\operatorname{curl} \mathbf{F} = \vec{\nabla} \times \mathbf{F} = \left\langle \frac{\partial}{\partial x}, \frac{\partial}{\partial y}, \frac{\partial}{\partial z} \right\rangle \times \langle F_x, F_y, F_z \rangle.$$

By the determinant definition of the cross product, this gives:

$$\vec{\nabla} \times \mathbf{F} = \begin{vmatrix} \hat{\imath} & \hat{\jmath} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ F_x & F_y & F_z \end{vmatrix} = \begin{pmatrix} \frac{\partial F_z}{\partial y} - \frac{\partial F_y}{\partial z} \\ \frac{\partial F_x}{\partial z} - \frac{\partial F_z}{\partial x} \\ \frac{\partial F_y}{\partial x} - \frac{\partial F_x}{\partial y} \end{pmatrix}$$

5.6 Surface and Volume Integrals

5.7 Stokes' Theorem

Stokes' Theorem

$$\iint_D \left(\vec{\nabla} \times \mathbf{F} \right) \cdot \hat{n} \, \mathrm{d}A = \oint_C \mathbf{F} \cdot \mathrm{d}\vec{r}$$

Generalized Stokes' Theorem

Let S be some n-dimensional orientable manifold and ∂S be the boundary of that manifold in $n - 1$ dimensions. Then:

$$\int_S \mathrm{d}\omega = \int_{\partial S} \omega.$$

This theorem generalizes the general theme of the fundamental theorems of calculus, which always go something along the lines of “the integral of some differential (“sum of little changes”) over some boundary is equal to the total change over the boundary.

5.8 Flux Integrals and the Divergence Theorem

Divergence Theorem

$$\iiint_V \left(\vec{\nabla} \cdot \mathbf{F}\right) \mathrm{d}V = \oiint_S \mathbf{F} \cdot \hat{n} \, \mathrm{d}A.$$